ROCSCIENCE

Case4_Iter1
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

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Slide2 Analysis Information

Case4_Iter1

Project Summary

File Name:	Case4_Iter1.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:01.400s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	30
Circles per division:	15
Number of iterations:	1
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	16 kN/m3
Cohesion	9 kPa
Phi	32 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	1.235900
Center:	20.735, 68.290
Radius:	44.257
Left Slip Surface Endpoint:	30.023, 25.018
Right Slip Surface Endpoint:	58.368, 45.000
Resisting Moment:	57529.4 kN-m
Driving Moment:	46548.4 kN-m
Total Slice Area:	115.99 m2
Surface Horizontal Width:	28.3454 m
Surface Average Height:	4.09204 m

Method: spencer

FS	1.233110
Center:	20.735, 68.290
Radius:	44.257
Left Slip Surface Endpoint:	30.023, 25.018
Right Slip Surface Endpoint:	58.368, 45.000
Resisting Moment:	57399.5 kN-m
Driving Moment:	46548.4 kN-m
Resisting Horizontal Force:	1053.21 kN
Driving Horizontal Force:	854.106 kN
Total Slice Area:	115.99 m2
Surface Horizontal Width:	28.3454 m
Surface Average Height:	4.09204 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	4536
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	3936
Number of Invalid Surfaces:	600

Error Codes
Error Code -108 reported for 73 surfaces
Error Code -111 reported for 527 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.2359

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.13382	5.87824	12.8671	Material 1	9	32	8.87806	10.9724	3.15649	0	3.15649	5.18448	5.18448
2	1.13382	17.3477	14.3777	Material 1	9	32	13.2948	16.4311	11.8923	0	11.8923	15.3003	15.3003
3	1.13382	28.237	15.8985	Material 1	9	32	17.372	21.4701	19.9564	0	19.9564	24.9044	24.9044
4	1.13382	38.5336	17.4309	Material 1	9	32	21.1136	26.0943	27.3567	0	27.3567	33.9858	33.9858
5	1.13382	48.2231	18.9763	Material 1	9	32	24.5227	30.3076	34.0992	0	34.0992	42.5317	42.5317
6	1.13382	57.2891	20.5362	Material 1	9	32	27.6012	34.1123	40.1882	0	40.1882	50.5277	50.5277
7	1.13382	65.7129	22.1122	Material 1	9	32	30.3504	37.5101	45.6257	0	45.6257	57.9573	57.9573
8	1.13382	73.4734	23.706	Material 1	9	32	32.7706	40.5012	50.4125	0	50.4125	64.8019	64.8019
9	1.13382	80.5469	25.3195	Material 1	9	32	34.8612	43.0849	54.5472	0	54.5472	71.0405	71.0405
10	1.13382	86.9062	26.9549	Material 1	9	32	36.6204	45.2591	58.0266	0	58.0266	76.6493	76.6493
11	1.13382	92.5207	28.6143	Material 1	9	32	38.0456	47.0206	60.8457	0	60.8457	81.6012	81.6012
12	1.13382	97.3553	30.3005	Material 1	9	32	39.1333	48.3649	62.997	0	62.997	85.8651	85.8651
13	1.13382	101.37	32.0162	Material 1	9	32	39.8788	49.2862	64.4713	0	64.4713	89.406	89.406
14	1.13382	104.519	33.7647	Material 1	9	32	40.2759	49.777	65.2569	0	65.2569	92.1833	92.1833
15	1.13382	106.749	35.5497	Material 1	9	32	40.3176	49.8285	65.3393	0	65.3393	94.1503	94.1503
16	1.13382	107.999	37.3754	Material 1	9	32	39.9951	49.4299	64.7013	0	64.7013	95.2526	95.2526
17	1.13382	108.196	39.2467	Material 1	9	32	39.2979	48.5683	63.3226	0	63.3226	95.4266	95.4266
18	1.13382	107.256	41.1695	Material 1	9	32	38.2143	47.229	61.1792	0	61.1792	94.5973	94.5973
19	1.13382	105.076	43.1505	Material 1	9	32	36.7298	45.3943	58.2431	0	58.2431	92.6749	92.6749
20	1.13382	101.535	45.1981	Material 1	9	32	34.828	43.0439	54.4816	0	54.4816	89.5513	89.5513
21	1.13382	96.48	47.3225	Material 1	9	32	32.4897	40.154	49.8567	0	49.8567	85.0932	85.0932
22	1.13382	89.7243	49.5363	Material 1	9	32	29.6926	36.6971	44.3247	0	44.3247	79.1348	79.1348
23	1.13382	73.2714	51.8556	Material 1	9	32	24.3072	30.0413	33.6731	0	33.6731	64.6238	64.6238
24	1.13382	45.87	54.3018	Material 1	9	32	16.2807	20.1213	17.7978	0	17.7978	40.4563	40.4563
25	1.13382	15.7785	56.9039	Material 1	9	32	8.0634	9.96556	1.54522	0	1.54522	13.9163	13.9163

Global Minimum Query (spencer) - Safety Factor: 1.23311

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.13382	5.87824	12.8671	Material 1	9	32	11.6987	14.4258	8.68307	0	8.68307	11.3554	11.3554
2	1.13382	17.3477	14.3777	Material 1	9	32	16.7286	20.6282	18.609	0	18.609	22.8972	22.8972
3	1.13382	28.237	15.8985	Material 1	9	32	21.1688	26.1034	27.3713	0	27.3713	33.4007	33.4007
4	1.13382	38.5336	17.4309	Material 1	9	32	25.059	30.9005	35.0481	0	35.0481	42.916	42.916
5	1.13382	48.2231	18.9763	Material 1	9	32	28.434	35.0622	41.7082	0	41.7082	51.4857	51.4857
6	1.13382	57.2891	20.5362	Material 1	9	32	31.3241	38.6261	47.4116	0	47.4116	59.1458	59.1458
7	1.13382	65.7129	22.1122	Material 1	9	32	33.7562	41.6251	52.211	0	52.211	65.9264	65.9264
8	1.13382	73.4734	23.706	Material 1	9	32	35.7537	44.0883	56.1531	0	56.1531	71.8523	71.8523
9	1.13382	80.5469	25.3195	Material 1	9	32	37.3375	46.0413	59.2786	0	59.2786	76.9435	76.9435
10	1.13382	86.9062	26.9549	Material 1	9	32	38.5258	47.5066	61.6234	0	61.6234	81.2151	81.2151
11	1.13382	92.5207	28.6143	Material 1	9	32	39.3345	48.5038	63.2193	0	63.2193	84.678	84.678
12	1.13382	97.3553	30.3005	Material 1	9	32	39.7778	49.0504	64.094	0	64.094	87.3387	87.3387
13	1.13382	101.37	32.0162	Material 1	9	32	39.868	49.1616	64.2719	0	64.2719	89.1999	89.1999
14	1.13382	104.519	33.7647	Material 1	9	32	39.6158	48.8507	63.7745	0	63.7745	90.2596	90.2596
15	1.13382	106.749	35.5497	Material 1	9	32	39.031	48.1295	62.6203	0	62.6203	90.5119	90.5119
16	1.13382	107.999	37.3754	Material 1	9	32	38.1216	47.0081	60.8256	0	60.8256	89.9458	89.9458
17	1.13382	108.196	39.2467	Material 1	9	32	36.8948	45.4954	58.4049	0	58.4049	88.5457	88.5457
18	1.13382	107.256	41.1695	Material 1	9	32	35.3572	43.5993	55.3705	0	55.3705	86.2901	86.2901
19	1.13382	105.076	43.1505	Material 1	9	32	33.5143	41.3268	51.7337	0	51.7337	83.1513	83.1513
20	1.13382	101.535	45.1981	Material 1	9	32	31.3711	38.684	47.5044	0	47.5044	79.0932	79.0932
21	1.13382	96.48	47.3225	Material 1	9	32	28.9326	35.6771	42.6923	0	42.6923	74.071	74.071
22	1.13382	89.7243	49.5363	Material 1	9	32	26.2036	32.3119	37.3069	0	37.3069	68.0267	68.0267
23	1.13382	73.2714	51.8556	Material 1	9	32	21.5609	26.587	28.1451	0	28.1451	55.5989	55.5989
24	1.13382	45.87	54.3018	Material 1	9	32	15.1104	18.6328	15.4158	0	15.4158	36.4456	36.4456
25	1.13382	15.7785	56.9039	Material 1	9	32	8.82355	10.8804	3.00929	0	3.00929	16.5466	16.5466

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.2359

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.0231	25.0184	0	0	0
2	31.1569	25.2774	9.24854	0	0
3	32.2907	25.5681	20.866	0	0
4	33.4245	25.891	34.1179	0	0
5	34.5583	26.247	48.3181	0	0
6	35.6921	26.6369	62.8276	0	0
7	36.826	27.0616	77.053	0	0
8	37.9598	27.5223	90.446	0	0
9	39.0936	28.0202	102.504	0	0
10	40.2274	28.5566	112.769	0	0
11	41.3612	29.1332	120.833	0	0
12	42.495	29.7517	126.333	0	0
13	43.6289	30.4143	128.964	0	0
14	44.7627	31.1232	128.473	0	0
15	45.8965	31.8812	124.673	0	0
16	47.0303	32.6914	117.446	0	0
17	48.1641	33.5575	106.755	0	0
18	49.2979	34.4838	92.6587	0	0
19	50.4318	35.4753	75.3264	0	0
20	51.5656	36.5382	55.0655	0	0
21	52.6994	37.6799	32.353	0	0
22	53.8332	38.9096	7.88265	0	0
23	54.967	40.2388	-17.3692	0	0
24	56.1008	41.6825	-38.4233	0	0
25	57.2347	43.2605	-48.0487	0	0
26	58.3685	45	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.23311

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.0231	25.0184	0	0	0
2	31.1569	25.2774	11.0146	6.99963	32.4354
3	32.2907	25.5681	24.5722	15.6152	32.4352
4	33.4245	25.891	39.7331	25.2498	32.4354
5	34.5583	26.247	55.6672	35.3756	32.4353
6	35.6921	26.6369	71.6432	45.5281	32.4353
7	36.826	27.0616	87.0198	55.2997	32.4353
8	37.9598	27.5223	101.239	64.3356	32.4352
9	39.0936	28.0202	113.819	72.3301	32.4353
10	40.2274	28.5566	124.352	79.0238	32.4353
11	41.3612	29.1332	132.5	84.2017	32.4353
12	42.495	29.7517	137.992	87.6916	32.4353
13	43.6289	30.4143	140.624	89.3643	32.4353
14	44.7627	31.1232	140.26	89.133	32.4353
15	45.8965	31.8812	136.833	86.9551	32.4353
16	47.0303	32.6914	130.348	82.834	32.4353
17	48.1641	33.5575	120.888	76.8222	32.4352
18	49.2979	34.4838	108.619	69.0259	32.4354
19	50.4318	35.4753	93.8053	59.6118	32.4353
20	51.5656	36.5382	76.8155	48.815	32.4353
21	52.6994	37.6799	58.1476	36.9519	32.4353
22	53.8332	38.9096	38.4526	24.436	32.4353
23	54.967	40.2388	18.5716	11.802	32.4354
24	56.1008	41.6825	2.38332	1.51456	32.4353
25	57.2347	43.2605	-4.81084	-3.05722	32.4354
26	58.3685	45	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	95, 20 95, 45 55, 45 30, 25 10, 25 10, 20